

ABSTRACT

Predicting NCAA March Madness Outcomes

The aim of this project is to develop a machine learning framework to predict outcomes for the NCAA Division I Mens Basketball Tournament. The purpose is to build upon traditional statistical rankings by implementing an automated pipeline that generates precise win probabilities for every possible tournament matchup. Extensive research into historical sports forecasting and modern machine learning techniques was conducted to establish a baseline for predictive accuracy and model calibration.

Following a comprehensive review of existing literature, the models were constructed using R. The primary dataset includes historical game results and Massey Ordinals, supplemented by custom feature engineering designed to capture team momentum and efficiency metrics. Four different modeling techniques were evaluated on the training data: Logistic Regression, Linear Regression, Random Forest, and XGBoost.

Unlike standard classification tasks, the final model was evaluated based on its ability to minimize the Brier Score, ensuring that the predicted probabilities were well calibrated against actual tournament results. The findings indicate that the integration of ensemble methods and strength of schedule adjustments provides a competitive edge in forecasting the high variance environment of the tournament. The complete codebase and dataset configurations are hosted on GitHub for further research and reproducibility. (do i put the link here or somewhere else)

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INTRODUCTION

The National Collegiate Athletic Association (NCAA) serves as the premier developmental system for the National Basketball Association (NBA) and the Women's National Basketball Association (WNBA). With thousands of athletes across both divisions arriving from around the world, the influence of the NCAA is extensive. As the basketball world has evolved into a multibillion dollar industry [1], the value of a competitive edge has become substantial. Teams historically relied on live scouting and film study to gain these advantages. However, the last decade has seen a major shift toward statistical findings. Inspired largely by the movie *Moneyball*, teams now look to analytics to uncover value that the naked eye might miss.

The focal point of this competitive landscape is March Madness. At the conclusion of the regular season, the selection committee invites 68 teams to participate in this tournament. The field includes the champions from all 32 Division I conferences alongside the best remaining teams chosen at large. The committee seeds these participants into four distinct regions with rankings ranging from 1 to 16. The four teams judged to be the strongest are awarded the number 1 seeds while the next tier receives the number 2 seeds. Within each region, the teams compete in a single elimination format where matchups are initially determined by rank. For example, the top seed plays the sixteenth seed. The winner of each region advances to the Final Four to compete for the national championship. Crucially, these contests take place at neutral sites rather than on home courts. This structure creates a predictive challenge where volatility is the norm.

To evaluate these predictive challenges in a rigorous and competitive environment, this project utilizes the Kaggle March Machine Learning Mania competition as its primary experimental framework. Kaggle provides a unique platform where data scientists from across the globe compete to minimize error in tournament forecasts using historical data stretching back to 1985. Unlike traditional bracket challenges that reward binary picks, this competition requires the submission of win probabilities for every possible matchup in the tournament field. This shift from simple selection to probabilistic estimation allows for a more granular analysis of model performance and calibration. By using this structured setting, the research can leverage standardized datasets including team efficiency ratings and historical results to build a pipeline that is both reproducible and statistically sound.

This statistical consistency is possible because basketball is uniquely suited for machine learning applications. Both individual and collective performances are captured through a standardized set of metrics. Discrete variables such as points, steals, assists, and rebounds provide a consistent numerical foundation for analysis across different eras and levels of competition. This uniformity allows for the creation of structured datasets where team strength can be quantified and compared through objective performance indicators. By leveraging these recurring statistical patterns, machine learning models can identify underlying trends that govern game outcomes and team efficiency.

A basic assessment often relies on a record of wins and losses to infer the overall quality of a team. While these records serve as strong indicators,

they frequently fail to capture critical nuance. Furthermore, the single elimination structure of the tournament ensures that upsets remain a constant possibility regardless of historical dominance. A defining example occurred in when the top ranked Virginia Cavaliers entered the postseason with 31 wins and only 3 losses. They fell to the lowest ranked UMBC Retrievers by 20 points. Such outcomes demonstrate why simple heuristics fail in this environment.

This inherent volatility drives a massive secondary market focused on sports betting. As legal wagering expands globally, March Madness has evolved into one of the largest gambling events in the world with 3.1 billion dollars wagered in 2025 on tournament outcomes [2]. To manage this risk, bookmakers generate odds which serve as the industry standard for predictive accuracy. These odds represent a market consensus that attempts to price in every available variable from player injuries to historical trends. However, these lines are often influenced by public sentiment and profit margins rather than pure probability. Consequently, developing a model capable of outperforming these market established baselines represents the gold standard for predictive success.

This paper aims to construct a predictive framework capable of meeting this standard. The primary objective is to develop and evaluate distinct statistical models that minimize the Brier Score metric used in the Kaggle competition. By testing various algorithms against historical data, the research seeks to identify a strategy that generates a winning score. The final analysis benchmarks these models against competitor submissions that utilized multiple strategies including elo ratings, standard odds conversion,

and analysis rankings. Ultimately, the optimal model will be deployed in the upcoming 2026 March Madness tournament to compare its projections with live results.

Motivated by this challenge, substantial research has addressed the complexities of March Madness prediction. The majority of this work focuses on engineering feature sets for learning algorithms ranging from linear regression to more complex ensemble methods. While the Kaggle competition provides foundational game statistics, transforming these into effective predictors requires significant effort. The following subsection details the specific data types provided by the competition to establish this baseline. The final subsection we review existing literature with a primary focus on feature engineering. Finally, the paper details the implementation of distinct modeling techniques. This comparative analysis examines linear and logistic regression alongside additional statistical methods. These models are compared against one another and evaluated relative to the winning scores from prior competitions.

Data Available

The highly quantifiable nature of basketball ensures that statistical data is both abundant and accessible. While the Kaggle competition permits participants to incorporate any external datasets they acquire, the organizers themselves provide a comprehensive archive. This repository includes over 30 distinct tables of historical data spanning more than three decades. All of the tables are separated by gender, with the only exception being the public rankings table. This is due to the fact that they only compiled mens rankings.

The data provided by Kaggle is split into regular season and postseason files. For the purposes of this project, the tournament phase is referred to as the playoffs. This research focuses primarily on the detailed results tables for both the regular season and the playoffs while also utilizing the essential rankings and seeding tables. Although the broader archive extends further back, the detailed game statistics required for this analysis commence in the 2003 season. Each entry in these tables provides a comprehensive statistical breakdown for both the winning and losing teams in every recorded matchup.

Season	DayNum	WTeamID	WScore	LTeamID	LScore	WLoc	NumOT	WFGM	WFGA	WFGM3	WFGA3	WFTM	WFTA	WDR	WDR	WAsst	WTO	WSR	WBlk	WPF	LFGM	LFGA	LFGM3	LFGA3	LFTM	LFTA	LOR	LDR
2003	10	1104	68	1328	62	N	0	27	58	3	14	11	18	14	24	13	23	7	1	22	22	53	2	10	16	22	10	22
2003	10	1272	79	1393	63	N	0	26	62	8	20	10	19	15	28	16	13	4	4	18	24	67	6	24	9	20	20	25
2003	11	1296	73	1437	61	N	0	24	58	8	18	17	29	17	26	15	10	5	2	25	22	73	3	26	14	23	31	22
2003	11	1296	56	1457	50	N	0	18	38	3	9	17	31	6	19	11	12	14	2	18	18	49	6	22	8	15	17	20
2003	11	1409	77	1208	71	N	0	30	61	6	14	11	13	17	22	12	14	4	4	20	24	62	6	16	17	27	21	15
2003	11	1458	81	1186	55	H	0	26	57	6	12	23	27	12	24	12	9	9	3	18	20	46	3	11	12	17	6	22
2003	12	1161	89	1236	62	H	0	23	55	2	8	32	39	13	18	14	17	11	1	25	19	41	4	15	20	28	9	21
2003	12	1186	75	1457	61	N	0	28	62	4	14	15	21	13	35	19	19	7	2	21	20	59	4	17	17	23	8	25
2003	12	1194	71	1156	66	N	0	28	58	5	11	10	18	9	22	9	17	9	2	23	24	52	6	18	12	27	13	26
2003	12	1458	84	1296	56	H	0	32	67	5	17	15	19	14	22	11	6	12	0	13	23	52	3	14	7	12	9	23
2003	13	1166	106	1426	59	H	0	41	69	15	25	9	13	15	29	21	11	10	6	16	17	52	4	11	12	17	8	15
2003	13	1202	74	1106	73	N	0	29	51	7	13	9	11	6	21	18	15	7	1	5	29	63	10	22	5	5	13	16
2003	13	1237	66	1135	65	N	0	26	66	5	19	9	13	21	23	15	17	12	3	17	24	56	6	19	11	17	14	21
2003	13	1323	76	1125	48	H	0	25	56	10	23	16	23	8	35	18	13	14	19	13	18	64	8	24	4	8	14	26
2003	14	1125	83	1135	77	N	1	30	70	11	31	12	15	15	18	22	16	18	5	21	28	69	4	15	17	27	20	27
2003	14	1156	78	1236	71	N	0	27	46	10	18	14	24	8	25	18	15	2	6	18	23	50	7	20	18	24	6	17
2003	14	1161	81	1194	56	H	0	22	48	5	12	32	43	16	32	13	24	5	3	19	21	65	2	13	12	19	16	18
2003	14	1186	82	1202	57	H	0	33	61	8	19	8	14	9	23	21	17	14	5	18	23	55	2	11	9	14	13	23
2003	14	1183	73	1129	59	A	0	29	59	9	19	6	8	8	21	17	15	10	1	19	21	51	3	14	14	18	12	22
2003	14	1314	85	1336	55	H	0	32	60	5	15	16	23	12	34	18	14	8	4	12	21	71	3	23	10	11	15	20
2003	14	1323	89	1237	45	H	0	34	62	8	16	13	21	12	34	25	11	9	13	16	18	74	4	15	5	11	27	22
2003	14	1353	69	1162	36	H	0	23	57	4	19	10	15	15	18	10	12	14	6	18	12	39	3	8	9	13	11	23
2003	14	1390	61	1131	57	H	0	20	53	6	27	15	24	15	24	9	14	4	2	17	19	54	3	17	16	18	11	22
2003	14	1426	59	1106	47	N	0	25	53	1	5	8	12	15	32	14	16	6	4	7	20	60	6	18	1	2	8	15
2003	14	1462	87	1389	48	H	0	35	70	6	19	11	16	17	27	18	7	13	3	10	20	63	7	20	1	4	16	18
2003	15	1161	77	1156	63	H	0	27	58	5	10	18	27	18	26	15	12	5	4	21	23	51	3	13	14	20	7	19
2003	15	1194	93	1236	86	N	2	33	75	9	20	18	33	17	26	14	14	17	0	21	27	58	14	31	18	22	9	33
2003	15	1196	76	1256	55	H	0	28	48	4	13	16	24	8	23	18	18	13	5	19	19	49	1	5	16	22	12	17
2003	15	1242	81	1221	57	H	0	28	53	3	5	22	37	13	29	12	19	10	3	21	18	52	7	21	14	21	8	22
2003	15	1422	84	1447	65	H	0	33	53	3	8	15	20	7	36	12	22	5	9	19	26	70	3	25	10	16	14	15
2003	16	1314	71	1353	67	H	0	27	57	5	14	12	19	14	30	19	20	8	5	13	27	66	9	22	4	11	12	19
2003	16	1390	63	1462	62	H	0	23	57	6	15	11	21	17	20	12	10	8	1	19	18	45	6	20	20	28	11	17
2003	17	1196	99	1183	65	H	0	39	73	11	28	10	22	16	31	32	14	10	14	19	22	66	4	11	17	24	15	23
2003	17	1304	68	1147	45	N	0	27	58	6	18	8	13	10	28	13	17	5	5	26	10	50	2	16	23	31	16	24
2003	18	1104	82	1186	56	H	0	24	48	10	20	24	34	12	26	13	13	7	3	14	19	55	8	21	10	12	11	18

Figure 1. Regular Season Detailed Data

This table records key performance metrics for each game played. The schema utilizes the prefix W to denote statistics for the winning team and the prefix L for the losing team. The specific recorded variables include:

- Points Scored
- The number of field goals made and attempted (FGM-A).
- The number of 3-point field goals made and attempted (FGM-A3).

- The number of free throws made and attempted (FTM-A).
- The number of offensive and defensive rebounds secured (OR- DR).
- The number of assists (Ast).
- The number of turnovers (TO).
- The number of blocks (Blk).
- The number of fouls (PF).

While raw box scores provide the foundation of events that happened in a game, they often fail to capture nuances in, and out, of the game. To address this limitation, this research supplements the Kaggle datasets with advanced metrics from analyst Ken Pomeroy (KenPom). Pomeroy is accepted as the industry standard for college basketball analysis. Pomeroy's approach is to normalize the data by calculating efficiency rather than raw stats. These statistics are updated daily throughout the regular season and tournament. For the purpose of this study, a snapshot of the rankings was acquired from the day prior to the start of the 2025 tournament.

2026 Pomeroy College Basketball Ratings													
help													
02-03-04-05-06-07-08-09-10-11-12-13-14-15-16-17-18-19-20-21-22-23-24-25-26													
Data through games of Monday, January 19 (3689 games)													
Rk	Team	Conf	W-L	NetRtg	ORTg	DRtg	AdjT	Luck	Strength of Schedule			NCSOS	
									NetRtg	ORTg	DRtg	NetRtg	
1	Michigan	B10	16-1	+36.00	127.0	91.0	73.5	+0.22	144	+11.23	116.6	105.4	+11.45
2	Arizona	B12	18-0	+34.04	126.9	92.9	71.8	+0.33	122	+6.59	114.8	108.2	+2.69
3	Duke	ACC	17-1	+32.78	126.0	93.2	67.6	+0.78	43	+9.69	115.7	106.0	+6.04
4	Purdue	B10	17-1	+31.31	129.5	98.2	65.9	+0.87	35	+9.41	116.1	106.7	+7.23
5	Houston	B12	17-1	+31.14	124.1	93.0	63.6	+0.54	78	+5.44	113.5	108.1	+0.85
6	Gonzaga	WCC	19-1	+31.04	126.3	95.2	70.3	+0.96	23	+5.48	113.1	107.7	+7.66
7	Illinois	B10	15-3	+30.44	129.3	98.9	66.4	+0.04	180	+8.62	115.5	106.9	+3.56
8	Iowa St.	B12	16-2	+30.00	125.0	95.0	68.2	+0.33	121	+5.29	112.3	107.0	-2.15
9	Florida	SEC	13-5	+29.57	125.6	96.1	70.4	-0.90	343	+12.33	117.9	105.6	+6.97
10	Michigan St.	B10	16-2	+29.32	120.0	90.7	66.3	-0.32	256	+8.60	116.7	108.1	+2.69
11	Connecticut	BE	18-1	+28.74	121.9	93.2	65.5	+0.87	34	+9.44	114.6	105.2	+7.69
12	Vanderbilt	SEC	16-2	+28.13	127.1	98.9	71.2	+0.32	123	+7.96	115.1	107.1	+1.72
13	Nebraska	B10	18-0	+27.06	123.5	96.5	67.3	+0.92	29	+5.94	113.8	107.9	-4.60
14	Virginia	ACC	16-2	+26.84	124.7	97.9	65.8	+0.02	185	+4.52	113.2	108.7	-3.64
15	BYU	B12	16-2	+26.70	125.2	98.5	70.8	+0.40	105	+7.19	113.9	106.8	+3.71
16	Louisville	ACC	13-5	+26.05	126.9	100.9	70.3	-0.54	300	+7.76	113.2	105.4	+2.11
17	Alabama	SEC	13-5	+25.46	129.4	104.0	73.8	-0.04	198	+14.83	119.1	104.2	+12.50
18	Iowa	B10	13-5	+25.37	123.0	97.7	63.9	-0.70	321	+4.71	112.3	107.6	-7.12
19	Kansas	B12	13-5	+25.25	122.0	96.8	67.8	-0.16	232	+12.99	116.4	103.4	+10.51
20	St. John's	BE	13-5	+24.44	123.1	98.7	70.9	-0.64	312	+9.28	115.8	106.5	+7.09

Figure 2. The first 20 teams of the KenPom data table.

The primary metrics utilized from this source include:

- Net Rating (NetRtg).
- Offensive Rating (ORTg).
- Defensive Rating (DRtg).
- Adjusted Tempo (AdjT).

Net rating is the differential between Offensive and Defensive Rating.

Offensive Rating measures the points scored per 100 possessions. Similarly, defensive rating measure the points given up per 100 possessions. These ratings are adjusted by the quality of opponents a team has played, and the game location. Lastly, Adjusted tempo is the number of possessions per 40 minutes. We define a possession as the period where a team controls the ball, ending with a shot attempt, turnover, or score.

While Pomeroy’s ratings serve as the industry standard, they are exclusively available for the Men’s division. Consequently, this research incorporates an additional dataset from Bart Torvik that focuses on the Women’s division. Torvik cites Pomeroy as a significant influence; thus, the ratings are constructed using a comparable methodology. However, distinct algorithmic differences exist between the two systems. For further detail regarding these distinctions, the reader is referred to [3]. The metrics used from Barttovik are:

- Adjusted Offensive Efficiency (ADJOE).
- Adjusted Defensive Efficiency (ADJDE).
- Adjusted Tempo (ADJ T.).

RK	TEAM	CONF	G	D1 AVG	PTS.2		PTS.1		PTS.0		PTS.3		PTS.4		PTS.5		PTS.6		PTS.7		PTS.8		PTS.9		PTS.10		PTS.11		PTS.12		PTS.13		PTS.14		PTS.15		PTS.16		PTS.17		PTS.18		PTS.19		PTS.20		PTS.21		PTS.22		PTS.23		PTS.24		PTS.25		PTS.26		PTS.27		PTS.28		PTS.29		PTS.30		PTS.31		PTS.32		PTS.33		PTS.34		PTS.35		PTS.36		PTS.37		PTS.38		PTS.39		PTS.40		PTS.41		PTS.42		PTS.43		PTS.44		PTS.45		PTS.46		PTS.47		PTS.48		PTS.49		PTS.50		PTS.51		PTS.52		PTS.53		PTS.54		PTS.55		PTS.56		PTS.57		PTS.58		PTS.59		PTS.60		PTS.61		PTS.62		PTS.63		PTS.64		PTS.65		PTS.66		PTS.67		PTS.68		PTS.69		PTS.70		PTS.71		PTS.72		PTS.73		PTS.74		PTS.75		PTS.76		PTS.77		PTS.78		PTS.79		PTS.80		PTS.81		PTS.82		PTS.83		PTS.84		PTS.85		PTS.86		PTS.87		PTS.88		PTS.89		PTS.90		PTS.91		PTS.92		PTS.93		PTS.94		PTS.95		PTS.96		PTS.97		PTS.98		PTS.99		PTS.100		PTS.101		PTS.102		PTS.103		PTS.104		PTS.105		PTS.106		PTS.107		PTS.108		PTS.109		PTS.110		PTS.111		PTS.112		PTS.113		PTS.114		PTS.115		PTS.116		PTS.117		PTS.118		PTS.119		PTS.120		PTS.121		PTS.122		PTS.123		PTS.124		PTS.125		PTS.126		PTS.127		PTS.128		PTS.129		PTS.130		PTS.131		PTS.132		PTS.133		PTS.134		PTS.135		PTS.136		PTS.137		PTS.138		PTS.139		PTS.140		PTS.141		PTS.142		PTS.143		PTS.144		PTS.145		PTS.146		PTS.147		PTS.148		PTS.149		PTS.150		PTS.151		PTS.152		PTS.153		PTS.154		PTS.155		PTS.156		PTS.157		PTS.158		PTS.159		PTS.160		PTS.161		PTS.162		PTS.163		PTS.164		PTS.165		PTS.166		PTS.167		PTS.168		PTS.169		PTS.170		PTS.171		PTS.172		PTS.173		PTS.174		PTS.175		PTS.176		PTS.177		PTS.178		PTS.179		PTS.180		PTS.181		PTS.182		PTS.183		PTS.184		PTS.185		PTS.186		PTS.187		PTS.188		PTS.189		PTS.190		PTS.191		PTS.192		PTS.193		PTS.194		PTS.195		PTS.196		PTS.197		PTS.198		PTS.199		PTS.200		PTS.201		PTS.202		PTS.203		PTS.204		PTS.205		PTS.206		PTS.207		PTS.208		PTS.209		PTS.210		PTS.211		PTS.212		PTS.213		PTS.214		PTS.215		PTS.216		PTS.217		PTS.218		PTS.219		PTS.220		PTS.221		PTS.222		PTS.223		PTS.224		PTS.225		PTS.226		PTS.227		PTS.228		PTS.229		PTS.230		PTS.231		PTS.232		PTS.233		PTS.234		PTS.235		PTS.236		PTS.237		PTS.238		PTS.239		PTS.240		PTS.241		PTS.242		PTS.243		PTS.244		PTS.245		PTS.246		PTS.247		PTS.248		PTS.249		PTS.250		PTS.251		PTS.252		PTS.253		PTS.254		PTS.255		PTS.256		PTS.257		PTS.258		PTS.259		PTS.260		PTS.261		PTS.262		PTS.263		PTS.264		PTS.265		PTS.266		PTS.267		PTS.268		PTS.269		PTS.270		PTS.271		PTS.272		PTS.273		PTS.274		PTS.275		PTS.276		PTS.277		PTS.278		PTS.279		PTS.280		PTS.281		PTS.282		PTS.283		PTS.284		PTS.285		PTS.286		PTS.287		PTS.288		PTS.289		PTS.290		PTS.291		PTS.292		PTS.293		PTS.294		PTS.295		PTS.296		PTS.297		PTS.298		PTS.299		PTS.300		PTS.301		PTS.302		PTS.303		PTS.304		PTS.305		PTS.306		PTS.307		PTS.308		PTS.309		PTS.310		PTS.311		PTS.312		PTS.313		PTS.314		PTS.315		PTS.316		PTS.317		PTS.318		PTS.319		PTS.320		PTS.321		PTS.322		PTS.323		PTS.324		PTS.325		PTS.326		PTS.327		PTS.328		PTS.329		PTS.330		PTS.331		PTS.332		PTS.333		PTS.334		PTS.335		PTS.336		PTS.337		PTS.338		PTS.339		PTS.340		PTS.341		PTS.342		PTS.343		PTS.344		PTS.345		PTS.346		PTS.347		PTS.348		PTS.349		PTS.350		PTS.351		PTS.352		PTS.353		PTS.354		PTS.355		PTS.356		PTS.357		PTS.358		PTS.359		PTS.360		PTS.361		PTS.362		PTS.363		PTS.364		PTS.365		PTS.366		PTS.367		PTS.368		PTS.369		PTS.370		PTS.371		PTS.372		PTS.373		PTS.374		PTS.375		PTS.376		PTS.377		PTS.378		PTS.379		PTS.380		PTS.381		PTS.382		PTS.383		PTS.384		PTS.385		PTS.386		PTS.387		PTS.388		PTS.389		PTS.390		PTS.391		PTS.392		PTS.393		PTS.394		PTS.395		PTS.396		PTS.397		PTS.398		PTS.399		PTS.400		PTS.401		PTS.402		PTS.403		PTS.404		PTS.405		PTS.406		PTS.407		PTS.408		PTS.409		PTS.410		PTS.411		PTS.412		PTS.413		PTS.414		PTS.415		PTS.416		PTS.417		PTS.418		PTS.419		PTS.420		PTS.421		PTS.422		PTS.423		PTS.424		PTS.425		PTS.426		PTS.427		PTS.428		PTS.429		PTS.430		PTS.431		PTS.432		PTS.433		PTS.434		PTS.435		PTS.436		PTS.437		PTS.438		PTS.439		PTS.440		PTS.441		PTS.442		PTS.443		PTS.444		PTS.445		PTS.446		PTS.447		PTS.448		PTS.449		PTS.450		PTS.451		PTS.452		PTS.453		PTS.454		PTS.455		PTS.456		PTS.457		PTS.458		PTS.459		PTS.460		PTS.461		PTS.462		PTS.463		PTS.464		PTS.465		PTS.466		PTS.467		PTS.468		PTS.469		PTS.470		PTS.471		PTS.472		PTS.473		PTS.474		PTS.475		PTS.476		PTS.477		PTS.478		PTS.479		PTS.480		PTS.481		PTS.482		PTS.483		PTS.484		PTS.485		PTS.486		PTS.487		PTS.488		PTS.489		PTS.490		PTS.491		PTS.492		PTS.493		PTS.494		PTS.495		PTS.496		PTS.497		PTS.498		PTS.499		PTS.500		PTS.501		PTS.502		PTS.503		PTS.504		PTS.505		PTS.506		PTS.507		PTS.508		PTS.509		PTS.510		PTS.511		PTS.512		PTS.513		PTS.514		PTS.515		PTS.516		PTS.517		PTS.518		PTS.519		PTS.520		PTS.521		PTS.522		PTS.523		PTS.524		PTS.525		PTS.526		PTS.527		PTS.528		PTS.529		PTS.530		PTS.531		PTS.532		PTS.533		PTS.534		PTS.535		PTS.536		PTS.537		PTS.538		PTS.539		PTS.540		PTS.541		PTS.542		PTS.543		PTS.544		PTS.545		PTS.546		PTS.547		PTS.548		PTS.549		PTS.550		PTS.551		PTS.552		PTS.553		PTS.554		PTS.555		PTS.556		PTS.557		PTS.558		PTS.559		PTS.560		PTS.561		PTS.562		PTS.563		PTS.564		PTS.565		PTS.566		PTS.567		PTS.568		PTS.569		PTS.570		PTS.571		PTS.572		PTS.573		PTS.574		PTS.575		PTS.576		PTS.577		PTS.578		PTS.579		PTS.580		PTS.581		PTS.582		PTS.583		PTS.584		PTS.585		PTS.586		PTS.587		PTS.588		PTS.589		PTS.590		PTS.591		PTS.592		PTS.593		PTS.594		PTS.595		PTS.596		PTS.597		PTS.598		PTS.599		PTS.600		PTS.601		PTS.602		PTS.603		PTS.604		PTS.605		PTS.606		PTS.607		PTS.608		PTS.609		PTS.610		PTS.611		PTS.612		PTS.613		PTS.614		PTS.615		PTS.616		PTS.617		PTS.618		PTS.619		PTS.620		PTS.621		PTS.622		PTS.623		PTS.624		PTS.625		PTS.626		PTS.627		PTS.628		PTS.629		PTS.630		PTS.631		PTS.632		PTS.633		PTS.634		PTS.635		PTS.636		PTS.637		PTS.638		PTS.639		PTS.640		PTS.641		PTS.642		PTS.643		PTS.644		PTS.645		PTS.646		PTS.647		PTS.648		PTS.649		PTS.650		PTS.651		PTS.652		PTS.653		PTS.654		PTS.655		PTS.656		PTS.657		PTS.658		PTS.659		PTS.660		PTS.661		PTS.662		PTS.663		PTS.664		PTS.665		PTS.666		PTS.667		PTS.668		PTS.669		PTS.670		PTS.671		PTS.672		PTS.673		PTS.674		PTS.675		PTS.676		PTS.677		PTS.678		PTS.679		PTS.680		PTS.681		PTS.682		PTS.683		PTS.684		PTS.685		PTS.686		PTS.687		PTS.688		PTS.689		PTS.690		PTS.691		PTS.692		PTS.693		PTS.694		PTS.695		PTS.696		PTS.697		PTS.698		PTS.699		PTS.700		PTS.701		PTS.702		PTS.703		PTS.704		PTS.705		PTS.706		PTS.707		PTS.708		PTS.709		PTS.710		PTS.711		PTS.712		PTS.713		PTS.714		PTS.715		PTS.716		PTS.717		PTS.718		PTS.719		PTS.720		PTS.721		PTS.722		PTS.723		PTS.724		PTS.725		PTS.726		PTS.727		PTS.728		PTS.729		PTS.730		PTS.731		PTS.732		PTS.733		PTS.734		PTS.735		PTS.736		PTS.737		PTS.738		PTS.739		PTS.740		PTS.741		PTS.742		PTS.743		PTS.744		PTS.745		PTS.746		PTS.747		PTS.748		PTS.749		PTS.750		PTS.751		PTS.752		PTS.753		PTS.754		PTS.755		PTS.756		PTS.757		PTS.758		PTS.759		PTS.760		PTS.761		PTS.762		PTS.763		PTS.764		PTS.765		PTS.766		PTS.767		PTS.768		PTS.769		PTS.770		PTS.771		PTS.772		PTS.773		PTS.774		PTS.775		PTS.776		PTS.777		PTS.778		PTS.779		PTS.780		PTS.781		PTS.782		PTS.783		PTS.784		PTS.785		PTS.786		PTS.787		PTS.788		PTS.789		PTS.790		PTS.791		PTS.792		PTS.793		PTS.794		PTS.795		PTS.796		PTS.797		PTS.798		PTS.799		PTS.800		PTS.801		PTS.802		PTS.803		PTS.804		PTS.805		PTS.806		PTS.807		PTS.808		PTS.809		PTS.810		PTS.811		PTS.812		PTS.813		PTS.814		PTS.815		PTS.816		PTS.817		PTS.818		PTS.819		PTS.820		PTS.821		PTS.822		PTS.823		PTS.824		PTS.825		PTS.826		PTS.827		PTS.828		PTS.829		PTS.830		PTS.831		PTS.832		PTS.833		PTS.834		PTS.835		PTS.836		PTS.837		PTS.838		PTS.839		PTS.840		PTS.841		PTS.842		PTS.843		PTS.844		PTS.845		PTS.846		PTS.847		PTS.848		PTS.849		PTS.850		PTS.851		PTS.852		PTS.853		PTS.854		PTS.855		PTS.856		PTS.857		PTS.858		PTS.859		PTS.860		PTS.861		PTS.862		PTS.863		PTS.864		PTS.865		PTS.866		PTS.867		PTS.868		PTS.869		PTS.870		PTS.871		PTS.872		PTS.873		PTS.874		PTS.875		PTS.876		PTS.877		PTS.878		PTS.879		PTS.880		PTS.881		PTS.882		PTS.883		PTS.884		PTS.885		PTS.886		PTS.887		PTS.888		PTS.889		PTS.890		PTS.891		PTS.892		PTS.893		PTS.894		PTS.895		PTS.896		PTS.897		PTS.898		PTS.899		PTS.900		PTS.901		PTS.902		PTS.903		PTS.904		PTS.905		PTS.906		PTS.907		PTS.908		PTS.909		PTS.910		PTS.911		PTS.912		PTS.913		PTS.914		PTS.915		PTS.916		PTS.917		PTS.918		PTS.919		PTS.920		PTS.921		PTS.922		PTS.923		PTS.924		PTS.925		PTS.926		PTS.927		PTS.928		PTS.929		PTS.930		PTS.931		PTS.932		PTS.933		PTS.934		PTS.935		PTS.936		PTS.937		PTS.938		PTS.939		PTS.940		PTS.941		PTS.942		PTS.943		PTS.944		PTS.945		PTS.946		PTS.947		PTS.948		PTS.949		PTS.950		PTS.951		PTS.952		PTS.953		PTS.954		PTS.955		PTS.956	
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Figure 3. The first 10 teams of the Barttorvik table

To complement the efficiency metrics provided by Pomeroy and Torvik, this research also incorporates the comprehensive ranking systems aggregated by Kenneth Massey. While the previously discussed datasets focus on raw predictive efficiency, the Massey Ordinals provide a consolidated view of team hierarchy. This dataset aggregates rankings from over 40 distinct

mathematical systems, pollsters, and algorithms. These rankings typically present data as an ordinal integer where the top team is assigned the value of 1. Incorporating this data allows the model to leverage a "wisdom of the crowd" approach by utilizing the consensus of the broader analytical community.

TEAM	SORT	DP	EBP	EMK	ENG	ESR	FAS	HAS	INC	JNG	KPK	LOG	MAS	MB	MGS
Arizona	>>	4	5	2	2	1	1	4	2	1	3	2	1	1	2
Michigan	>>	1	2	1	1	6	3	1	1	4	1	1	6	6	4
Duke	>>	2	1	3	7	4	4	2	3	2	2	3	2	5	5
Purdue	>>	7	6	5	3	5	6	3	5	6	4	5	4	4	3
Gonzaga	>>	3	3	9	9	7	7	7	6	9	5	4	5	7	1
Connecticut	>>	13	11	6	5	3	5	15	7	5	6	7	7	3	6
Houston	>>	5	4	4	6	8	9	8	4	7	8	12	3	8	9

Figure 4. A sample of the Massey Ordinals table showing distinct ranking systems.

One significant challenge in aggregating these distinct sources is the inconsistency in team nomenclature. For instance, a university might be listed as "St. Mary's" in one dataset and "Saint Mary's CA" in another. To resolve this discrepancy, all external data was strictly mapped to the unique TeamID integer provided by the Kaggle competition files. This standardization ensures that the advanced efficiency metrics are accurately merged with the box score data for every specific matchup.

LITERATURE REVIEW

Past Strategies

The pursuit of a perfect bracket has generated a substantial body of research in predictive modeling over the last decade. While specific algorithms vary, the most successful strategies in the Kaggle Machine Learning Mania competition consistently rely on one of two distinct methodological frameworks.

The primary objective for this study is to minimize the Brier Score. We define the Brier Score (BS) as follows:

$$(1) \quad BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2$$

where N is the number of events, f_t is the predicted probability, and o_t is the actual outcome. For this research, a 0 is considered a loss, and a 1 is a win. A Brier Score of 0 would reflect perfect accuracy, while a Brier Score of 1 reflects perfect inaccuracy. To provide a baseline, a strategy where we predict every team has a 50% chance of winning would result in a Brier Score of 0.25. (thinking of putting another example here)

The first framework relies on fundamental analysis where the objective is to construct a ground-truth probability independent of public sentiment. A common implementation of this approach involves utilizing domain-specific heuristics such as the elo rating system. Originally designed for chess, elo provides a dynamic measure of team strength that updates after every game. It offers a robust baseline that accounts for strength of schedule more

effectively than raw win-loss records [10]. Many strategies also incorporate expert features. These include the Nate Silver, FiveThirtyEight, and KenPom ratings which act as high-signal inputs that aggregate player health, travel distance, and roster depth into a single numeric value. By treating these expert outputs as features rather than final predictions, modelers can leverage human domain expertise while correcting for known expert biases.

The second dominant framework relies on the Efficient Market Hypothesis [8]. This approach posits that global betting markets, specifically closing lines from major bookmakers, offer a more accurate aggregate predictor of game outcomes than individual statistical models. The efficacy of this strategy is well-documented in recent Kaggle competitions where five of the top eight performing models utilized market-derived features as their primary feature. These strategies do not attempt to beat the market. Instead, they focus on mathematically disentangling the true win probability from the bookmaker’s overround and inherent biases.

To achieve this, recent state-of-the-art implementations utilize specific conversion algorithms to de-noise betting data. The standard tool in this domain is the *Goto_conversion*. Competitors employ these functions to address the favorite-longshot bias. This is the tendency for markets to overvalue underdogs and undervalue favorites. By applying these corrections, modelers can transform raw Vegas odds into highly calibrated probabilities that minimize Brier Score more effectively than fundamental analysis alone. This approach effectively leverages the crowd wisdom of the market while stripping away the pricing inefficiencies introduced by public betting behavior. For more information on how this strategy works, we refer the reader to [9].

The applicability of these methodological frameworks, however, is heavily constrained by the structural discontinuity introduced by the 2021 Name, Image, and Likeness (NIL) legislation. Statistical analysis of the post-NIL landscape reveals a significant deviation from historical norms, characterized by a rapid consolidation of talent where the number of programs with elite win percentages ($> 75\%$) has nearly tripled [6]. This phenomenon implies that feature weights derived from pre-2021 eras are likely subject to significant concept drift. Consequently, this literature review restricts its focus to winning strategies from recent Kaggle competitions (2022-present), as methodologies developed before the NIL era may not reflect the approaches best suited to the heightened stratification of team strength driven by current transfer portal dynamics.

Modeling Techniques

Recent tournament prediction methodologies employ several established modeling and computational techniques. This section provides formal definitions for the five core approaches used in this study: Logistic Regression, Linear Regression, Monte Carlo Simulation, Random Forest, and Neural Networks.

Logistic Regression is a binary classification model that estimates the probability of an outcome by applying the logistic (also known as sigmoid) function to a linear combination of input features. The model is defined as:

$$(2) \quad P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

where Y is the binary outcome, $X = (x_1, x_2, \dots, x_n)$ represents the feature vector, and $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through the maximum likelihood estimation. This function ensures the predicted probabilities are bounded between $[0, 1]$. Logistic regression assumes a linear relationship between the input features and the log-odds of the outcome. This assumption can be restrictive when the true relationships are nonlinear. To address this limitation, common approaches are feature engineering with polynomial or interaction terms (e.g., $x^2, x_i \cdot x_j$) to capture nonlinear patterns.

Linear Regression is a supervised learning method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. Linear Regression

is defined as:

$$(3) \quad \hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where $\hat{Y}$ is the predicted continuous outcome, $X = (x_1, x_2, \dots, x_n)$ represents the vector of features, and $\beta = (\beta_0, \beta_1, \dots, \beta_n)$ are the model coefficients estimated through ordinary least squares (OLS) minimization. For more information on OLS we refer the reader to [11]. Unlike logistic regression, linear regression produces unbounded predictions and does not inherently output probabilities. In sports prediction contexts, linear regression is commonly used to estimate point totals, scoring margins, or other continuous performance metrics that can subsequently be transformed into win probabilities through additional techniques.

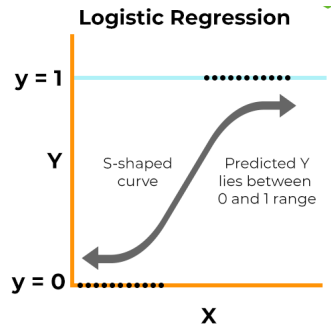


Figure 5. An Example of a Logistic Regression Equation [12].

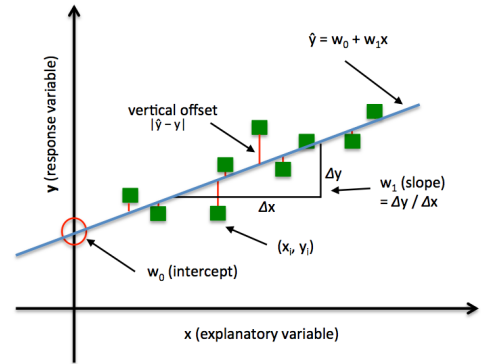


Figure 6. An Example of a Linear Regression Equation [13].

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate probabilities for systems with inherent uncertainty. The method generates N independent samples $x_1, x_2, \dots, x_N$

from a probability distribution $f(x)$, then estimates probabilities by computation the proportion of samples satisfying a certain condition. The simulation is defined as:

$$(4) \quad \hat{P} = \frac{1}{N} \sum_{i=1}^N 1(x_i \in A)$$

where $1(\cdot)$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. A represents the event of interest, and $\hat{P}$ is the Monte Carlo estimate of the true probability $P(A)$. By the Law of Large Numbers [14], $\hat{P} \rightarrow P(A)$ as $N \rightarrow \infty$. In sports prediction contexts, Monte Carlo methods are frequently employed to simulate game outcomes by sampling from estimated score distributions, enabling the estimation of win probabilities through empirical frequency counts across simulated matchups.

XGBoost (Extreme Gradient Boosting), considered a "black-box" approach, is an ensemble learning method that constructs a sequence of decision trees where each successive tree is trained to correct the residual errors of the previous trees. Unlike Random Forest, which builds trees independently, XGBoost employs gradient boosting to iteratively minimize a loss function through additive modeling [5]. The model prediction is defined as the sum of predictions from all trees:

$$(5) \quad \hat{y}_i = \sum_{k=1}^K f_k(x_i).$$

Here, $f_k(x_i)$ is the prediction of the k^{th} tree for the i^{th} data point, $\hat{y}_i$ is the final predicted value for the i^{th} data point, and K is the number of trees.

Each tree is trained to minimize the objective function

$$(6) \quad \text{obj}(\theta) = \sum_i^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where l is the loss function which simply computes the difference between the true and predicted, and Ω is the regularization term, discouraging overly complex trees. The optimization process then proceeds via gradient descent on the residuals. For more information regarding gradient descent we refer the reader to [4]. XGBoost is widely adopted in competitive machine learning for its performance on structured data and ability to capture complex interactions and relationships between features.

$$(7) \quad h^{(\ell)} = \sigma(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$$

if i do neural nets i will keep going with this.

Probability Prediction Models

This section details the development and evaluation of models designed to predict a continuous win probability $P \in (0, 1)$ for any given matchup. While the tournament results are ultimately binary, the Brier Score evaluation metric heavily penalizes overconfidence in incorrect predictions and rewards well-calibrated probabilities. Consequently, this research moves beyond simple classification to focus on models that quantify the uncertainty inherent in the March Madness environment. By establishing a framework where a value $P > 0.5$ denotes a predicted victory for Team A, a baseline is created for performance that accounts for the high variance of single-elimination play. This probabilistic approach serves as the engine of the project and bridges the gap between theoretical literature and the empirical application of machine learning.

To expand further on these concepts, the methodology begins with a rigorous data processing phase where historical box scores are transformed into a symmetric training set. This ensures that the models learn to evaluate the relative strength between two competitors rather than relying on arbitrary formatting artifacts. This section details the complete machine learning pipeline implemented for the 2026 tournament prediction. We first examine the data cleaning procedures required to tailor the feature sets to the specific statistical assumptions of each algorithm. Following this, we explore the variable selection process used to identify high signal metrics such as KenPom and Massey data. Finally, we implement and compare two distinct modeling architectures, specifically Logistic Regression and XGBoost, to determine

which approach best minimizes the Brier Score in a high variance environment.

Logistic Regression

Variable Selection

To accurately model basketball outcomes, one must first deconstruct the sport into its atomic statistical events. A standard game is composed of a finite sequence of possessions. Within these possessions, specific discrete actions determine the flow of the contest. The primary events recorded in the raw dataset include Field Goals, Free Throws, Rebounds, Turnovers, and Personal Fouls. A Field Goal (*FGM*) represents the primary method of scoring and is worth either two or three points. A Free Throw (*FTM*) is an unopposed attempt worth one point awarded after a foul. A Rebound is recorded when a player retrieves the ball after a missed shot; this is further categorized into Offensive Rebounds (*OR*), which grant the shooting team a second opportunity, and Defensive Rebounds (*DR*), which terminate the opponent's possession. Finally, a Turnover (*TO*) is a loss of possession due to a violation or steal, resulting in zero points.

Analyzing these events using raw totals is statistically flawed due to the confounding variable of pace. A team that plays at a high tempo may record high totals for points and rebounds simply because they generate more possessions, not because they are efficient. To resolve this, this model utilizes the "Four Factors of Basketball Success." Developed by Dean Oliver, this framework isolates the four specific components that correlate most strongly with winning: shooting efficiency, turnover avoidance, rebounding dominance, and free throw frequency [16].

Crucial to these efficiency calculations is the estimation of total Possessions ($Poss$). A possession is defined as the period during which a team controls the ball, ending with a shot attempt, a turnover, or a trip to the free-throw line. Since possessions are not explicitly tracked in standard box scores, they must be estimated using the following formula:

$$(8) \quad Poss = (FGA - OR) + TO + (0.44 \cdot FTA).$$

This equation aggregates the three ways a possession can end: a field goal attempt (FGA), a turnover (TO), or free throws (FTA). The term $(FGA - OR)$ accounts for missed shots that are rebounded by the offense; because the team retains the ball, the possession continues, and thus the initial attempt is subtracted to avoid double-counting. The coefficient 0.44 applied to Free Throw Attempts estimates the proportion of trips to the line that actually terminate a possession (e.g., the second of two shots), accounting for and-one (fouled while successfully scoring) situations and technical fouls. This derived metric serves as the denominator for rate statistics, allowing for a direct comparison between teams of differing tempos.

The first factor is Effective Field Goal Percentage ($eFG\%$). This metric refines standard field goal percentage by accounting for the fact that a three-point shot is worth 50% more than a two-point shot. It is calculated as:

$$(9) \quad eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA}$$

By including the 0.5 weighting for three-pointers ($FGM3$), this variable measures a team's ability to maximize points per attempt regardless of their shot selection strategy. The second factor is Turnover Percentage ($TOV\%$). Unlike raw turnover counts, this metric normalizes errors against the total number of possessions calculated above.

$$(10) \quad TOV\% = \frac{TO}{Poss}$$

This ratio identifies teams that value the basketball. A lower percentage indicates a disciplined offense that rarely forfeits scoring opportunities. The third factor is Offensive Rebound Percentage ($ORB\%$). This metric measures a team's physical dominance relative to their available opportunities. Using raw offensive rebound totals is misleading because a team that misses many shots will naturally have more chances to rebound. This metric corrects for that bias by comparing offensive rebounds (OR) to the total number of rebounds available at that specific rim:

$$(11) \quad ORB\% = \frac{OR}{OR + DR_{opp}}$$

Here, DR_{opp} represents the defensive rebounds secured by the opponent. A higher percentage indicates a team that frequently generates "second chance" points. Finally, the model incorporates Win Percentage ($Win\%$) as a fifth feature to capture intangible qualities such as coaching and clutch performance. This is calculated as the mean of the binary game results (1 for a win, 0 for a loss) over the course of the season. Crucially, the logistic

regression algorithm does not process these values in isolation. Instead, the dataset is transformed to calculate the differential between the two competing teams for every metric (e.g., $eFG\%_{diff} = eFG\%_{TeamA} - eFG\%_{TeamB}$). This centers the analysis on the relative mismatch between the opponents rather than their absolute statistics.

Logistic Regression

Boosting

Results

Here are my insights...

Point Spread Models

same as above

S

Here are my insights...

Comparison of models

My Contributions and Analysis

Here are my insights...

2026 Competition

Strategies

Here are my insights...

Results

CONCLUSIONS

Summary

As a preliminary assessment...

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